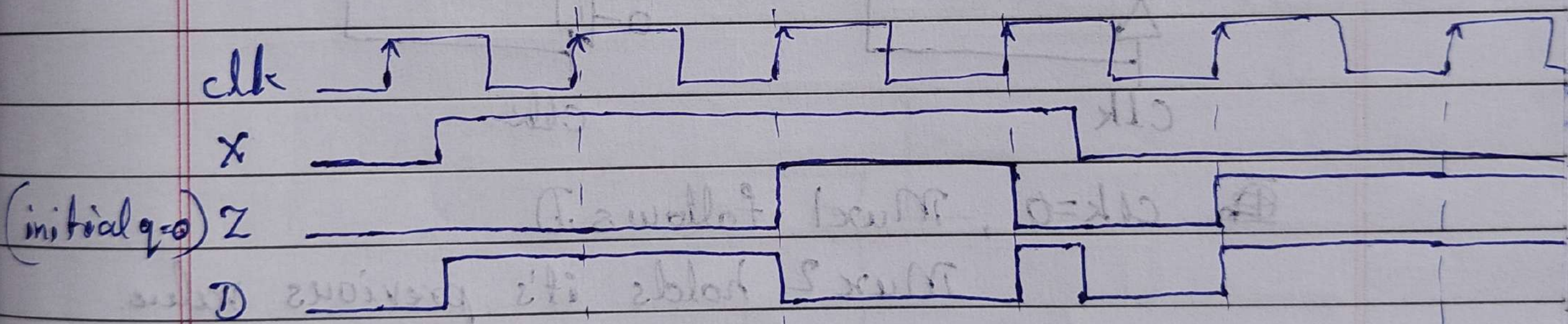
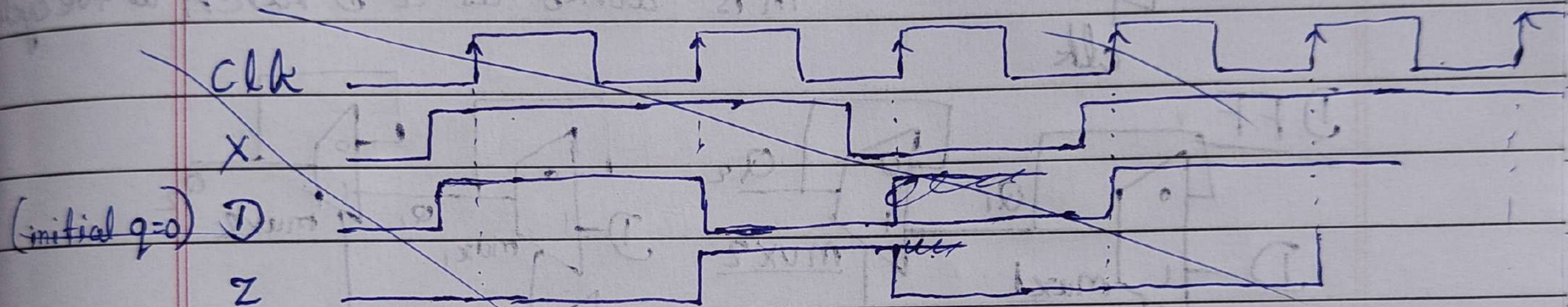
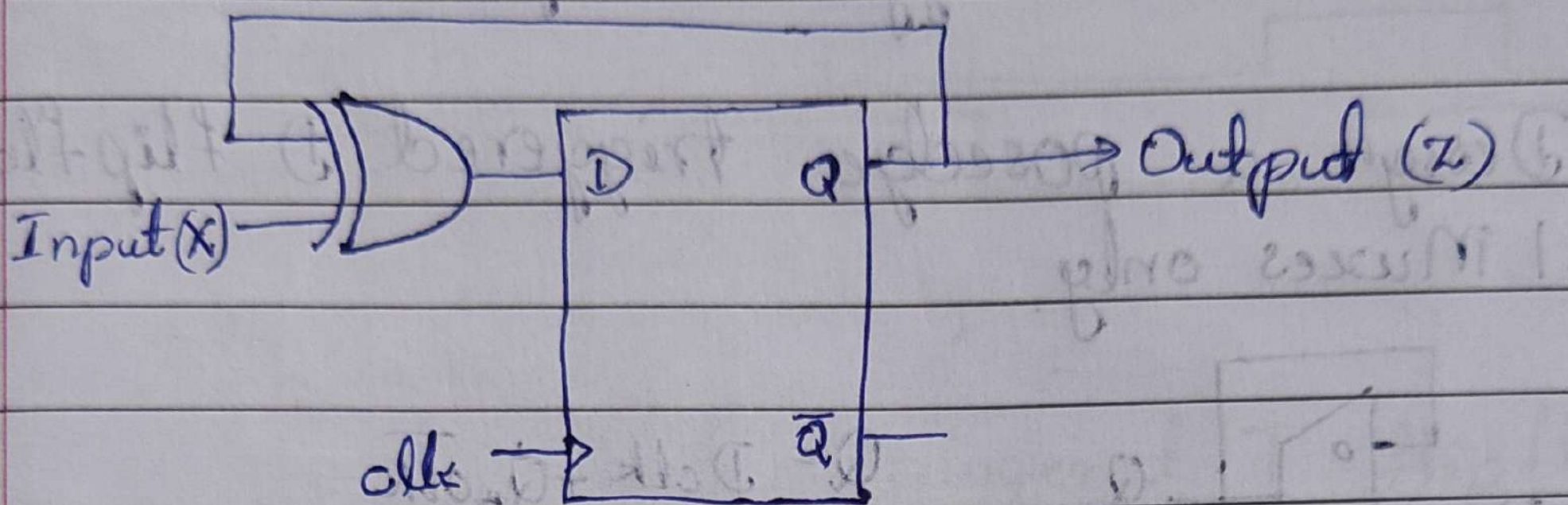


1. Explain the behavior of the given circuit.



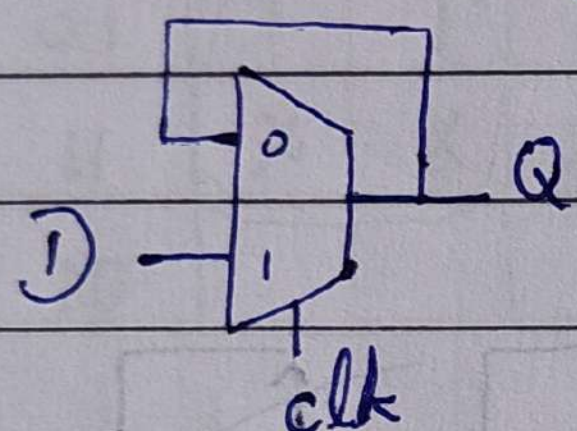
$$D = X \oplus Q$$

when $X=0$, ~~FF~~ XOR gate behaves like a buffer, $D=Q$. Therefore the DFF holds its current state. previous state.

when $X=1$, XOR gate behaves like a NOT gate, $D=\bar{Q}$. Therefore the DFF toggles its previous state.

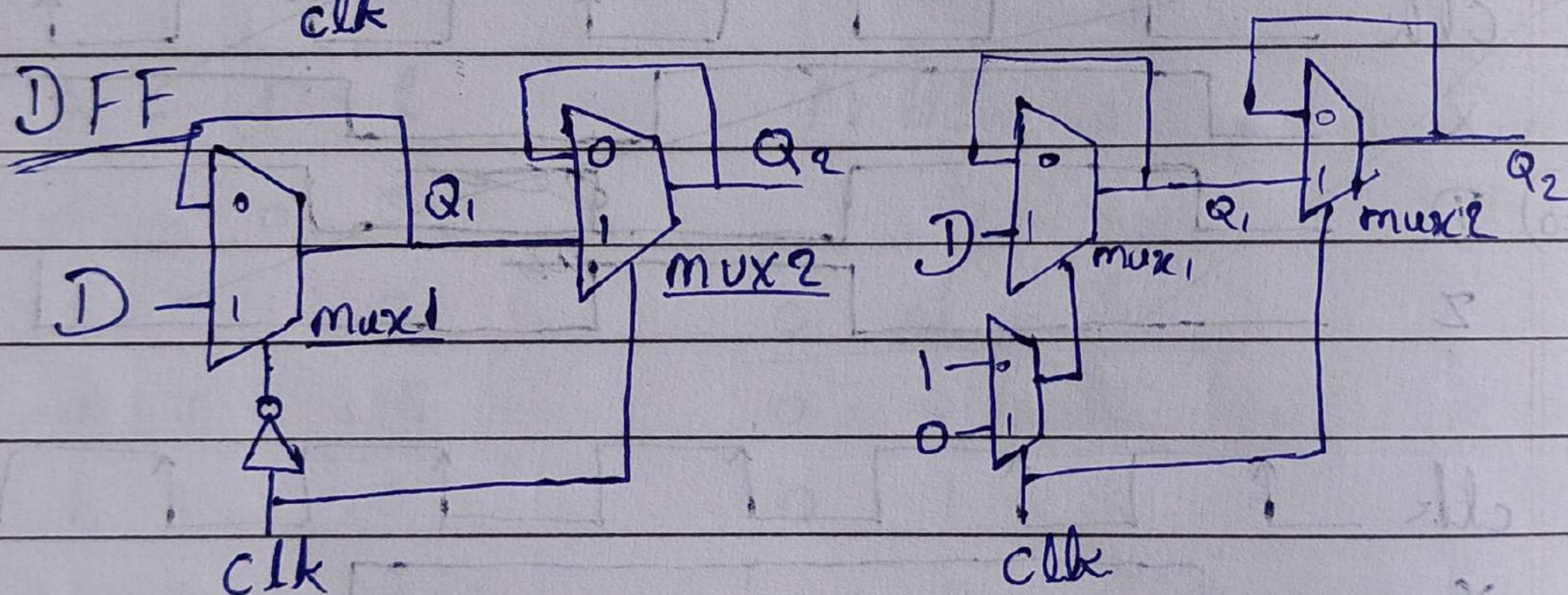
Hence, the circuit behaves like a TFF with X as a Toggle (T) input.

2. Design a posedge triggered D flipflop using 2:1 Muxes only.



$$Q = D \cdot \text{clk} + Q \cdot \bar{\text{clk}}$$

This works as a D latch at +ve cycle



When $\text{clk} = 0$, Mux1 follows D.

Mux2 holds its previous value.

$\text{clk} 0 \rightarrow 1$, Mux1 holds its previous value. and

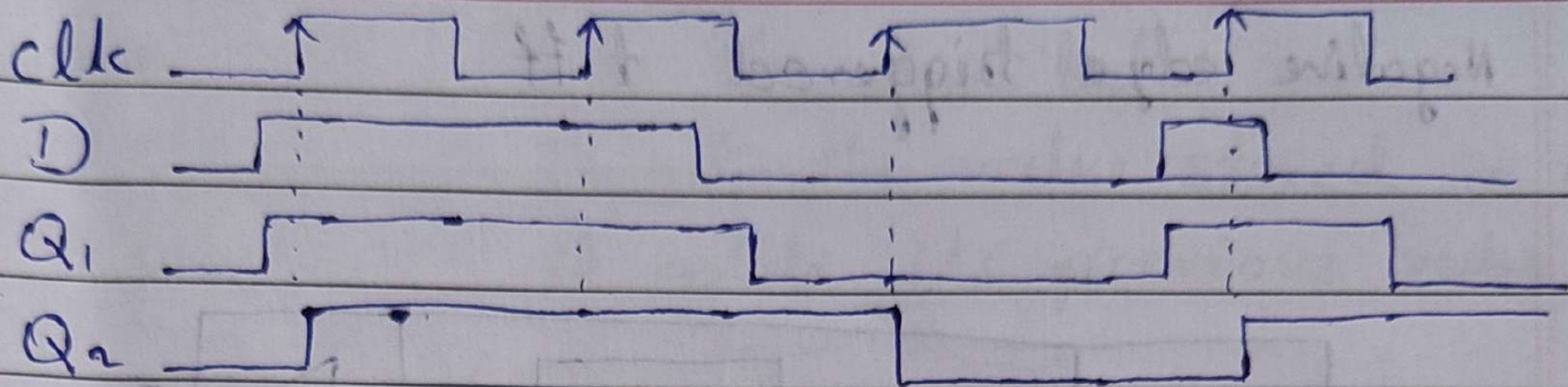
(Q_2 changes) $\rightarrow$ Mux2 follows the mux1's previous value.

$\text{clk} = 1 \rightarrow 0$, Mux1 starts to follow D, but

(Q_2 holds) $\rightarrow$ Mux2 holds the previous output.

The output of Mux2 behaves like

The output changes only on the rising edge of the clk, $\therefore$ The circuit acts like D flipflop.

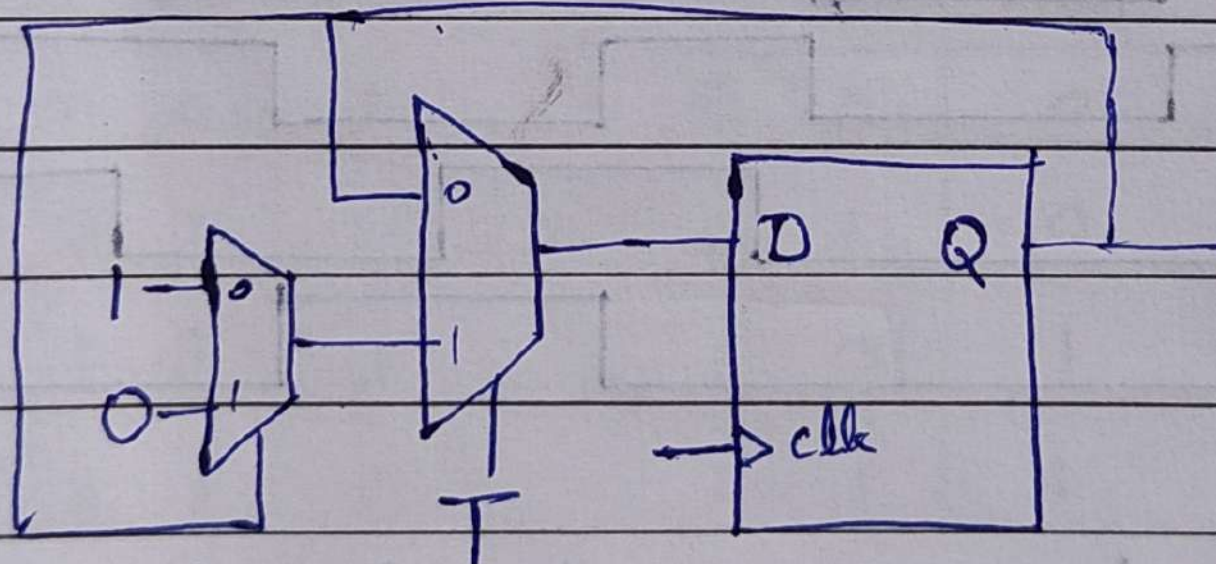


3. Design a negedge triggered T flip flop using 2:1 Muxes only.

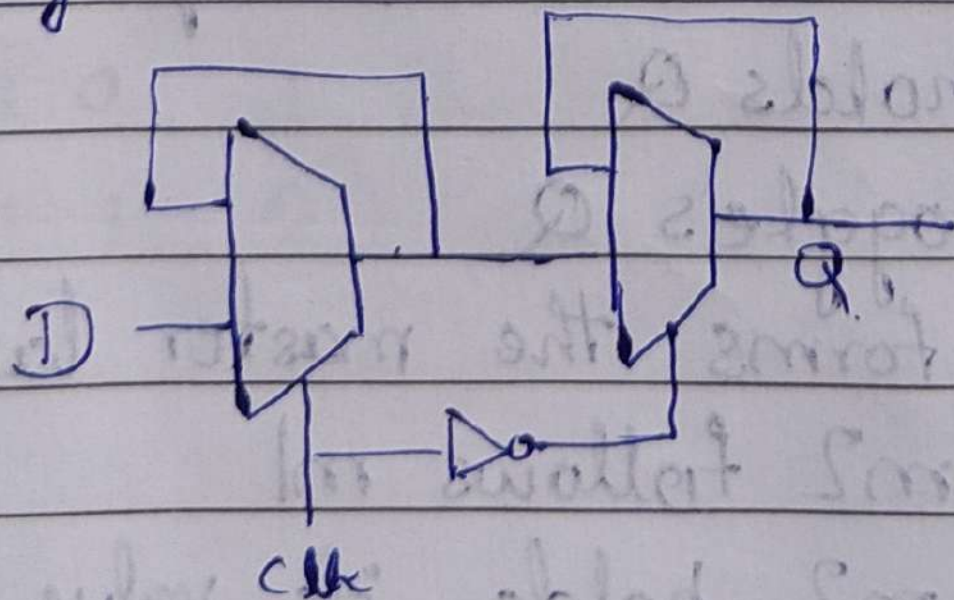
$$Q_{n+1} = D$$

$$Q_{n+1} = T \oplus Q_n$$

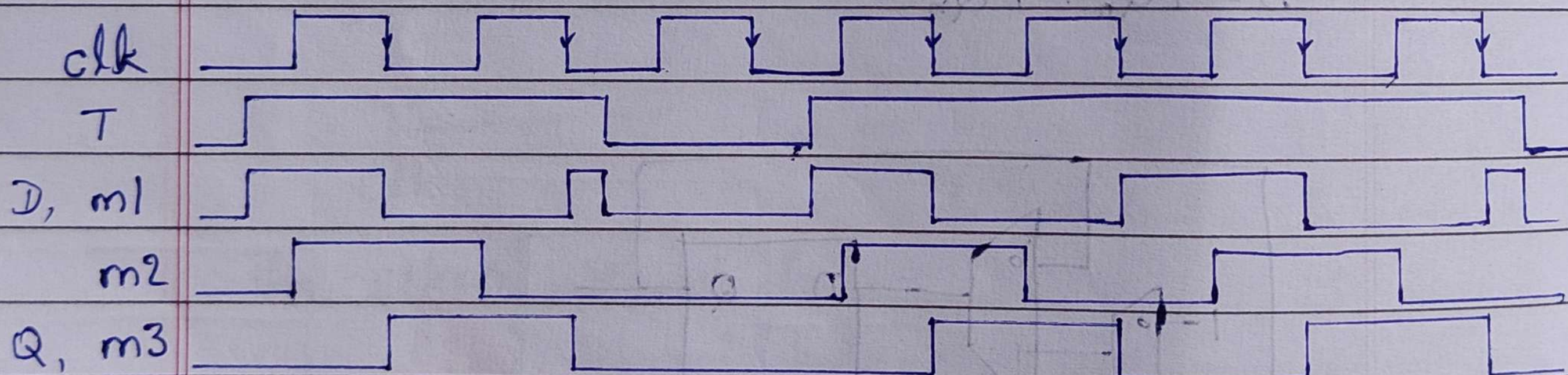
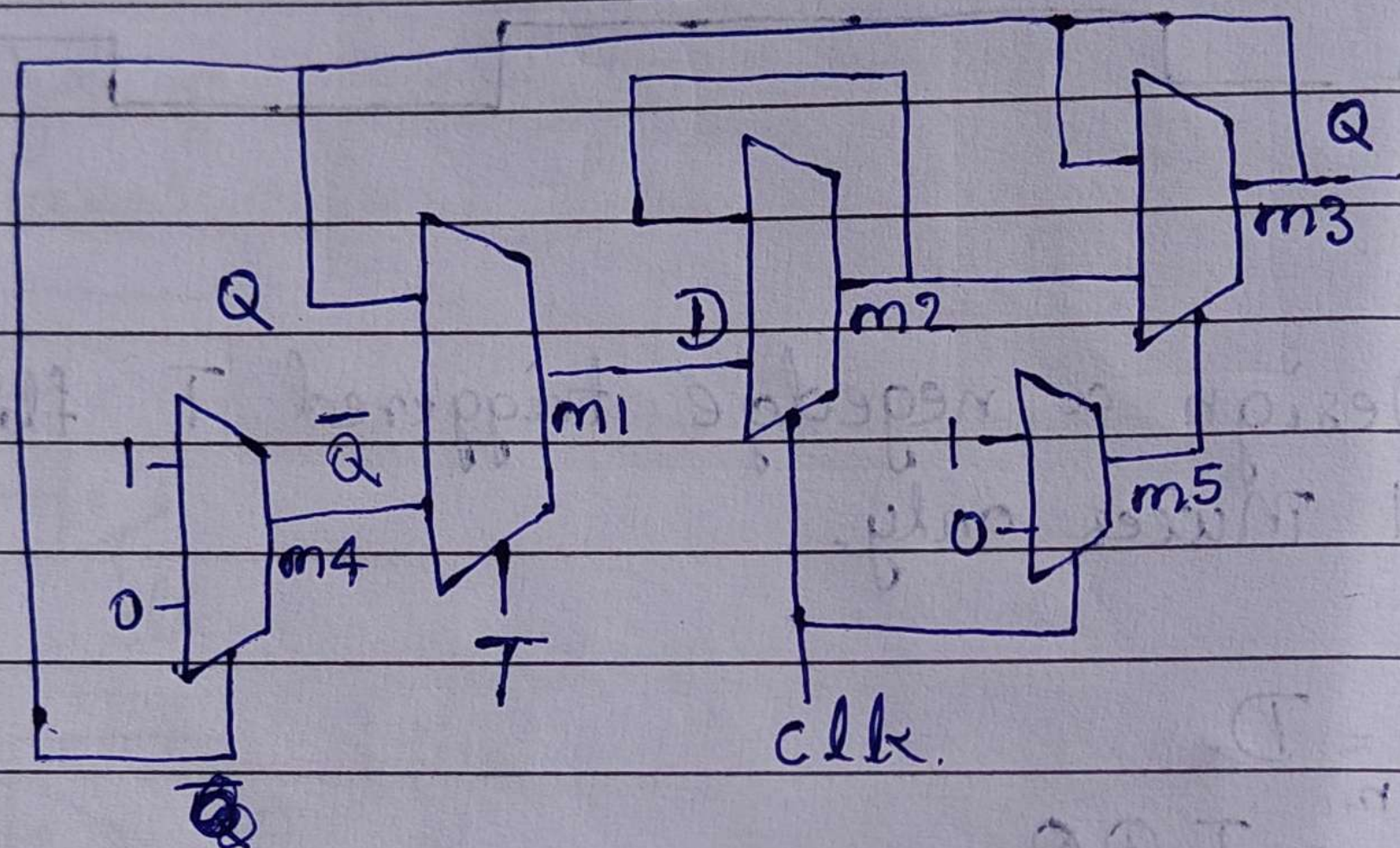
$$\therefore D = T\bar{Q}_n + \bar{T}Q_n$$



negedge D ff



Negative edge triggered tff



Mux m4 acts as a Not gate for Q, $m4 = \bar{Q}$.

Mux m1 acts as a T flip-flop logic

- $T=0 \rightarrow$ holds Q
- $T=1 \rightarrow$ toggles Q

Mux m2 ~~not~~ forms the master latch:

- $clk=1 \rightarrow$ m2 follows m1
- $clk=0 \rightarrow$ m2 holds its value.

mux m3 forms the slave latch:

$clk=0 \rightarrow Q$ gets the value stored in m2.

$clk=1 \rightarrow Q$ holds its previous value.

~~Hence when~~

Operation:

$T=0 \rightarrow$ output remains unchanged.

$T=1 \rightarrow$ output toggles at every negedge of clk .

$$Q_{n+1} = T \oplus Q_n$$

4. Convert by adding external gates:

a) A D flip-flop to a J-K flip-flop.

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

$Q_{n+1} (JK)$

J	Q_n		Q_{n+1}
	00	01	10
0	0	1	0
1	1	1	1

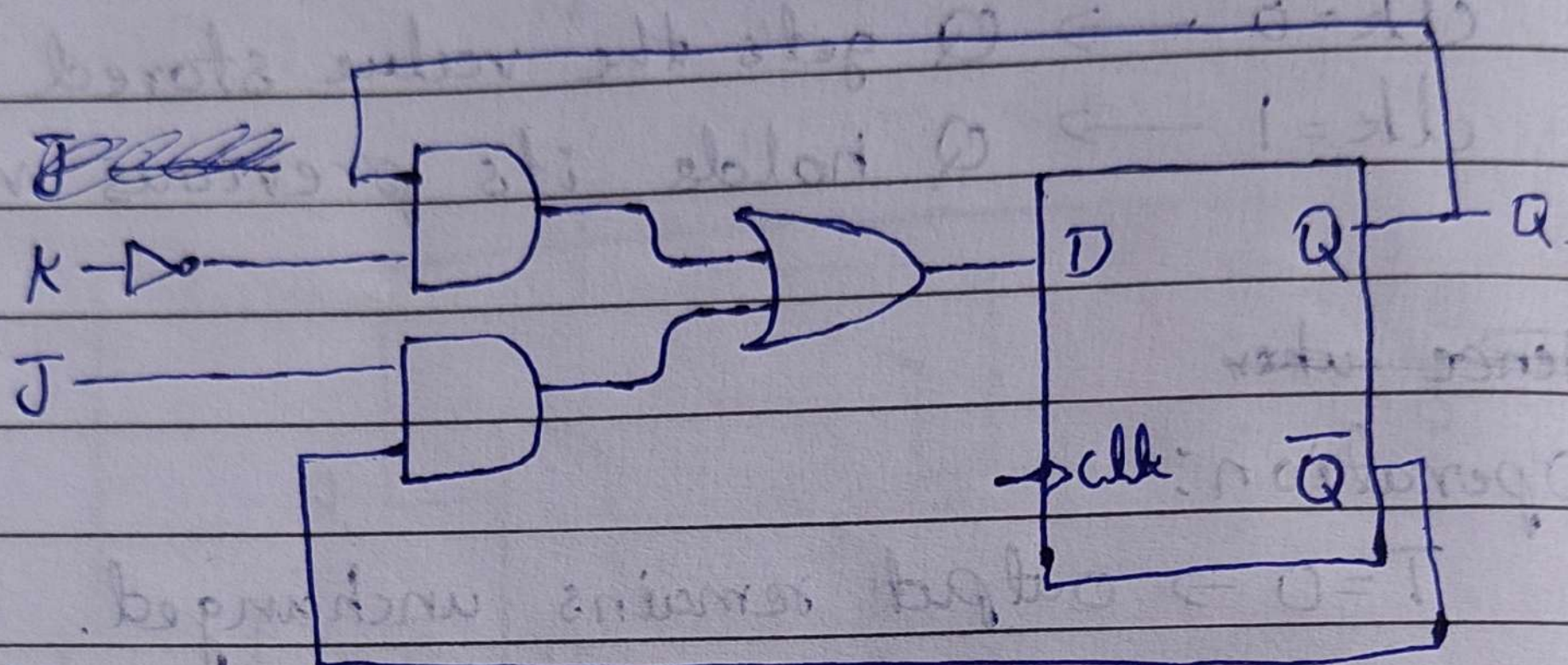
$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

$Q_{n+1} (D)$

D	Q_n		Q_{n+1}
	0	1	
0	0	0	0
1	1	1	1

$$Q_{n+1} = D$$

$$\therefore D = \bar{J}\bar{Q} + KQ$$



b) A T flip-flop to a D flip-flop.

T	Q _n	Q _{n+1}
---	----------------	------------------

0 0 0

0 1 1

1 0 1

1 1 0

D	Q _n	Q _{n+1}
---	----------------	------------------

0 0 0

0 1 0

1 0 1

1 1 1

$$Q_{n+1} = T \oplus Q_n$$

$$Q_{n+1} = D$$

$$D = T \oplus Q_n$$

$$D = \bar{T}\bar{Q}_n + TQ_n$$

D	Q _n	T
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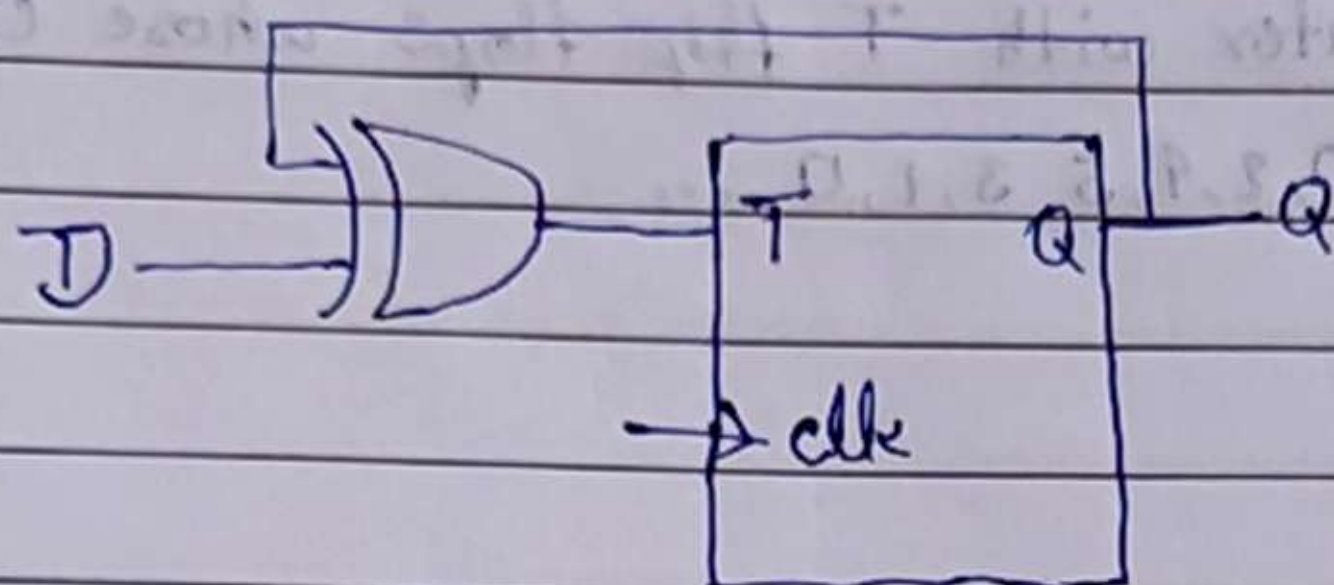
0 0 0

0 1 1

1 0 1

1 1 0

$$T = D \oplus Q$$



5. In a positive edge triggered JK flip-flop, we have $\bar{J} = \bar{Q}$ and $K = 1$ as shown in the figure below.

Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles.

$\bar{J}$	K	Q	$\bar{Q}$	
		0	1	← initially
1	1	1	0	1 st clock cycle
0	1	0	1	2 nd clock cycle
1	1	1	0	3 rd clock cycle
0	1	0	1	4 th clock cycle
1	1	1	0	5 th clock cycle
0	1	0	1	6 th clock cycle

sequence at Q: 0, 1, 0, 1, 0, 1.